Chinyere E. Ugwuanyi

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EDUCATION

Rutgers University – MBS, Analytics & Data Science | GPA: 3.89 | *May 2026*

- **Relevant Courses**: Regression, BI & Visual Analytics, Data Warehousing, Data Management, Data Structures (Java)

West Virginia State University – B.S. Chemistry | GPA: 3.84 | *April 2021*

TECHNICAL SKILLS

- **Programming & Tools:** Python (Pandas, NumPy, Scikit-learn), R, SQL/MySQL, SPSS, Qualtrics, Tableau, Excel
- **ML & Statistics:** Regression, Classification, Survival Analysis, Reliability Testing, Experimental Design, Longitudinal and Cohort Analysis
- **Data Engineering & Validation:** Data Cleaning, Feature Engineering, Model Validation, Quality Assurance

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | *Rutgers - Office of Institutional Research & Decision Support (OIRDS)* *March 2025 – Present*

- Developed an M-index dining utilization algorithm (n = **4,800+**, $p < 0.001$) supporting evidence-based resource allocation.
- Managed university-wide survey deployment (n = **40,000+**) across three campuses using Qualtrics distribution logic and response tracking.
- Performed large-scale roster QA using R, Python, SPSS, and Excel to ensure cross-campus data integrity.
- Conducted **longitudinal cohort analyses** on engagement, retention, and equity gaps to inform institutional planning.

Student Researcher - Computer Science Education | *Rutgers Women in Computer Science* *Spring 2024 - Fall 2024*

- Designed Qualtrics surveys (n = **900+**) to assess Computer Science and IT&I curriculum effectiveness.
- Built Tableau dashboards and visual reports supporting curriculum improvement recommendations.

PROJECTS

Rutgers University Student Organizations Survey Data Integration & Analysis | *Rutgers - Office of Institutional Research & Decision Support (OIRDS)*

- Integrated survey responses with official university club records by scraping institutional API data (941 organizations) and merging with respondent-provided text entries.
- Developed data-cleaning and fuzzy-matching pipelines in R to standardize club names, resolve duplicates, and link unstructured survey inputs to official organization identifiers.
- Produced a clean, analysis-ready dataset to support survey reporting, incentives distribution, and student engagement initiatives across multiple campuses.

ZIP-Code-Level COVID-19 Early Detection & Surveillance Model (NJ) *(In Progress)*

- Engineer dynamic spread indicators (velocity, acceleration, jerk, rolling variance, volatility ratios, correlation) from daily ZIP-code-level COVID-19 case data to capture real-time transmission dynamics.
- Apply drift detection (KS test, CUSUM, Isolation Forest) to identify structural shifts and emerging outbreaks earlier than traditional county-level models; benchmark performance against established epidemiological approaches to improve targeted intervention strategies.

WiDS Datathon 2026 - Wildfire Survival Modeling Project *(In Progress)*

- Build discrete-time survival model predicting when wildfires will reach within 5km of evacuation zones; (risk horizons: 12h, 24h, 48h, 72h) to support emergency-response logistics.
- Expand events to person-period data, engineer time encodings, train gradient-boosted hazard model, evaluate with cross-validated Brier scores.
- Produce calibrated, time-specific risk curves to inform evacuation timing, resource staging, and operational planning under uncertainty.

Fake News Categorization – ML Pipeline (~45,000 articles) | *Accenture* GitHub Repo

- Built a text classification pipeline using TF-IDF, cosine similarity, and Nearest Neighbors to detect misinformation and reduce training bias and data leakage.
- Engineered 20+ linguistic features (e.g., punctuation density, sentence structure, readability scores) supporting interpretable modeling.

Airbnb NYC Price Prediction Model – Applied ML GitHub Repo

- Built and validated regression model (40K+ listings, 5-fold CV), identifying key drivers of NYC price variability.

CERTIFICATIONS / PRESENTATIONS / ADDITIONAL INFORMATION

- **American Educational Research Association (AERA) Annual Meeting** (2026) – Co-author
  From Survey to Success: Predicting First-Year Retention Through Longitudinal Data Analysis
- **Rutgers University – Office of Institutional Research & Decision Support** - New Brunswick, NJ | *December 2025*
  Meet the Cohort of 2025: What Multi-Year New Student Survey Data Reveals About Our Newest Scarlet Knights
- **Cornell Tech** | **Break Through Tech AI/ML Program** - Machine Learning Foundations | *July 2025*
- **CITI Program** | **Social / Behavioral / Epidemiologic Research Investigators** | *June 2024*
- **LinkedIn Learning** – Business Analysis Foundations: Business Process Modeling, Data Analytics for Business Professionals, Turning Data into Strategies | *January 2026*